Reasons:

1. The frequency of crossing over is directly proportional to the relative distance of the genes on the chromosomes.
2. More crossing over = More distance between two genes and
   Less crossing over = Less distance between the two genes.

In the above problem, the sequence of the genes on the linkage map is B, A, C

NOTE: Teachers can give different crossing over percentage between its linked genes in a chromosome and make the students construct the linkage maps. The external examiner can also do the same for the Board Practical Examinations.

V - Experiments

Exercise 19: To dissect and display the pollinia of *Calotropis*

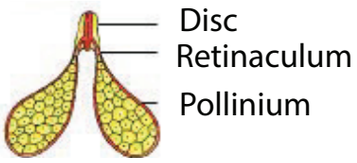


Figure 19: Pollinia of *Calotropis*

Aim:

To dissect and observe the structure of pollinia and understand the mechanism of pollination in *Calotropis* flowers. (Translator Mechanism)

Principle:

In *Calotropis* the pollen in each anther lobe of a stamen unites into a mass, forming a pollinium.

Requirements:

Flowers of *Calotropis*, dissection needle, dissection microscope ,slide, blade, glycerine, coverslip, scissors.

Procedure:

Take a mature flower of *Calotropis*. Observe the parts of the flower and remove the calyx and corolla with the help of scissors. Identify the pentangular stigmatic disc. Insert the needle at the angles of the stigma where the pollinia are adhered. Dissect it and place the pollinia on a clean slide. Mount it in glycerine and place a coverslip on it. Observe the pollinia under the dissection microscope and record your observation.

Observation:

The stamens of *Calotropis* produce pollinium. Two pollinia are found attached to a glandular adhesive disc called corpusculum by a thread like structure called retinaculum. The whole structure looks like inverted letter 'Y' and is called translator. The sticky disc gets attached with the legs of pollinator (bees or butterflies) and is carried to the stigma of another flower, thus ensuring pollination.

Inference:

The structure of pollinia of *Calotropis* is well suited to achieve pollination.

Exercise 20: Study of Pollen germination on a slide

NOTE: Pollen germination can be studied by dusting some pollens from common flowers like *Crotalaria*, *Hibiscus*, *Pisum*, etc. on a glass slide containing a drop of 10% sugar solution or tender coconut water or any nutrient medium.

Observe the slide after about 10 – 15 minutes under the low power of compound microscope. You will be able to observe the pollen tubes coming out of the pollen grains.

Aim:

To study the pollen germination on a slide.

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